

KOSUMPHISAI, Thailand

WMO: 483820

Lat: 16.245N

Lon: 103.069E

Elev: 154

StdP: 99.48

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.7	16.2	10.0	7.8	18.8	11.5	8.6	19.1	6.5	25.0	6.1	25.1	1.1	40	0.333

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.2	38.5	26.3	37.4	26.3	36.3	26.2	28.2	34.6	27.8	33.9	27.5	33.3	2.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	22.8	30.9	26.5	22.4	30.6	26.1	21.9	30.2	92.4	34.8	90.3	34.1	88.6	33.4	33.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.4	4.3	DB	12.3	39.9	1.5	1.2	11.2	40.7	10.3	41.4	9.5	42.1	8.4	42.9	(4)
(5)			WB	10.7	29.9	1.6	1.4	9.6	30.9	8.6	31.8	7.7	32.6	6.5	33.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	24.2	26.8	29.1	30.8	30.3	30.0	29.2	28.7	28.5	28.0	26.7	24.6
	DBStd	2.89	2.54	2.89	2.89	2.27	1.67	1.36	1.48	1.20	1.14	1.31	2.08	2.61
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6592	442	469	591	623	628	600	596	581	554	557	501	452
	CDD18.3	3552	184	236	333	373	369	350	337	322	304	299	251	193
	CDH23.3	40885	1637	2559	4152	4988	4687	4460	3999	3631	3332	3195	2479	1766
	CDH26.7	18921	640	1247	2235	2822	2381	2163	1746	1418	1252	1274	1050	692
Wind	WSAvg	1.3	1.4	1.2	1.3	1.2	1.2	1.5	1.6	1.5	1.0	1.0	1.3	1.4
Precipitation	PrecAvg	1300	9	16	45	87	169	168	193	236	261	95	18	3
	PrecMax	1696	74	66	131	245	350	300	338	349	421	220	110	21
	PrecMin	995	0	0	1	21	45	65	23	65	70	11	0	0
	PrecStd	171	16	17	39	58	74	63	78	67	88	51	25	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.6	37.4	39.5	40.4	39.4	37.2	36.8	35.0	34.5	34.4	35.2	34.6
		MCWB	23.7	24.6	25.0	26.9	27.8	27.1	26.7	27.2	27.3	26.7	25.8	25.1
	2%	DB	33.1	35.9	38.0	39.0	37.4	35.7	35.1	33.8	33.5	33.4	34.0	33.1
		MCWB	23.8	24.7	25.4	26.4	27.1	27.0	26.7	27.0	27.1	26.3	25.5	24.4
	5%	DB	31.7	34.6	36.7	37.6	35.9	34.8	33.9	33.0	32.7	32.6	32.9	31.8
		MCWB	23.5	24.7	25.6	26.6	27.0	26.8	26.5	26.7	26.8	25.8	25.0	23.7
	10%	DB	30.3	33.2	35.3	36.2	34.6	33.8	32.8	32.1	31.8	31.9	31.8	30.5
		MCWB	22.7	24.3	25.5	26.4	26.9	26.7	26.4	26.5	26.6	25.5	24.6	23.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.0	27.1	28.0	28.7	29.0	28.6	28.3	28.1	28.2	27.7	27.4	26.5
		MCDB	32.1	33.8	35.7	37.1	36.7	34.3	33.8	33.1	32.7	32.6	33.6	32.8
	2%	WB	25.2	26.3	27.1	28.0	28.2	27.9	27.4	27.5	27.7	27.1	26.6	25.4
		MCDB	30.8	33.0	34.7	36.1	35.0	33.5	32.8	32.4	32.1	31.9	32.2	31.1
	5%	WB	24.3	25.6	26.4	27.4	27.6	27.4	27.0	27.1	27.2	26.7	25.9	24.5
		MCDB	29.7	32.3	34.1	35.3	33.9	32.8	32.2	31.8	31.4	31.2	31.1	30.2
	10%	WB	23.5	25.0	25.9	26.8	27.2	27.1	26.6	26.7	26.9	26.2	25.2	23.7
		MCDB	28.8	31.5	33.6	34.5	33.1	32.3	31.5	31.1	30.9	30.4	29.9	29.3
Mean Daily Temperature Range	5% DB	MDBR	12.0	12.1	11.0	10.2	8.7	7.7	7.1	6.7	6.7	7.7	9.9	11.5
		MCDBR	13.2	13.2	12.7	11.7	10.3	9.0	8.7	8.0	7.9	8.5	10.8	12.0
		MCWBR	5.5	4.6	3.9	3.3	3.2	2.6	2.6	2.8	2.9	2.9	3.6	4.6
	5% WB	MCDBR	11.6	11.6	11.3	10.9	9.3	8.2	7.8	7.5	7.4	7.9	9.7	11.2
		MCWBR	5.8	4.8	4.4	3.7	3.4	2.9	2.7	2.9	3.0	3.1	3.9	4.9
Clear-Sky Solar Irradiance	taub	0.479	0.592	0.658	0.618	0.513	0.465	0.453	0.462	0.505	0.535	0.477	0.471	
	taud	2.074	1.796	1.688	1.821	2.139	2.294	2.326	2.292	2.157	2.036	2.170	2.121	
	Ebn at Noon	805	731	698	729	799	829	840	841	807	769	803	800	
	Edh at Noon	158	218	249	219	157	133	129	135	154	169	143	147	
All-Sky Solar Radiation	RadAvg	4.89	5.19	5.30	5.72	5.54	5.19	4.86	4.74	4.63	4.97	4.92	4.74	
	RadStd	0.29	0.25	0.25	0.24	0.30	0.26	0.34	0.29	0.32	0.35	0.18	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)		+0.29	N/A	N/A	N/A	+0.16	+0.13	N/A	N/A	N/A	+106

Nomenclature: See separate page